Discussion: Did the Results Match Our Expectation?

Model	Positional Encoding	Top-1 Test Acc
Original ViT	1D Learned (Additive)	67.86%
Modified ViT	2D RoPE (Rotational)	72.96%
Difference	**(Modified − Original)**	**+5.10%**

Yes, the results matched and exceeded our expectation. We expected 2D RoPE to perform comparably or slightly better than 1D learned embeddings. The Modified ViT outperformed the Original by 5.10%, which is a clear and meaningful improvement on CIFAR-100 (100 classes).

Why did 2D RoPE outperform?

The improvement can be attributed to three factors that aligned with our hypothesis:

First, the explicit 2D spatial encoding (row + column) gave the model a structural inductive bias that is especially helpful when training from scratch on only 45,000 images. The original ViT must learn spatial relationships implicitly from a flattened 1D sequence, which requires more data to converge.

Second, the per-layer positional signal from RoPE maintained spatial coherence across all 12 transformer layers. In the original ViT, the single additive positional embedding at the input may have its signal diluted as it passes through repeated self-attention and MLP layers.

Third, the relative position encoding allowed the model to learn translation-equivariant features, which is a natural fit for image recognition where the same object can appear at different locations.

Observations from Training Curves:

The loss curves show that 2D RoPE achieves consistently lower training and validation loss from approximately epoch 20 onward. The validation accuracy gap widens steadily and stabilizes at around 5%. Both models converge smoothly under the cosine learning rate schedule with warmup, confirming that the improvement comes from the positional encoding itself, not from training instability.

Conclusion:

2D RoPE is a viable and effective alternative to learned positional embeddings for ViTs. Its relative position encoding, explicit 2D structure, and per-layer application provide meaningful benefits when training from scratch on small datasets. The +5.10% test accuracy improvement demonstrates that positional encoding design is an important factor in ViT performance, particularly in data-limited regimes.